EXIONAICS: ENTERPRISES LEVEL EMISSION ESTIMATION DATASET WITH LARGE LANGUAGE MODELS

Yanming Guo & Jin Ma
University of Sydney
{yguo0337, j.ma}@sydney.edu.au

Xiao Qian & Kevin Credit
Maynooth Unievristy
{Qian.Xiao,Kevin.Credit}@mu.ie

ABSTRACT

Accurate greenhouse gas emission reporting is increasingly important for governments, businesses, and investors. However, mainstream adoption—particularly among small and medium enterprises—remains limited by the high implementation costs, fragmented emission factor databases, and a lack of robust classification tools. To address these challenges, we introduce **ExioNAICS**, the first large-scale NLP benchmark dataset for enterprise-level GHG emission estimation. ExioNAICS integrates validated North American Industry Classification System labels for over 20,850 companies with a concordance to an economic model of carbon intensity factors. By framing the classification task as an Information Retrieval problem and fine-tuning Sentence-BERT with a contrastive learning approach, we achieve state-of-the-art performance on NAICS categories, notably 77.51% Top-1 accuracy and 91.33% Top-10 accuracy in our most challenging setting 1,114 classes. We make ExioNAICS publicly available to lower the entry barrier for GHG reporting and facilitate broader collaboration between machine learning researchers and climate experts. Dataset, code and trained models could be found: `https://huggingface.co/datasets/Yvnminc/ExioNAICS`

1 INTRODUCTION

Accurate greenhouse gas (GHG) emission reporting has become increasingly critical for governments, businesses, and investors. Worldwide, governments are implementing mandatory GHG reporting programs to enhance transparency and accountability in emission reductions, with the European Union, Australia, and other regions mandating detailed enterprise-level disclosures (Holding, 2023; Heinkel et al., 2001; Chen & Zhang, 2024; Li et al., 2022). These regulations are based on the GHG Protocol Corporate Standard, which defines emissions across three scopes: Scope 1 direct emissions, Scope 2 indirect emissions from electricity consumption, and Scope 3 all other indirect emissions across the value chain (Hertwich & Wood, 2018). Among these, Scope 3 emissions are particularly challenging to estimate, as they involve emissions from upstream and downstream activities such as purchased goods, transportation, and waste disposal.

A fundamental issue in accurate GHG reporting lies in sectoral classification, which is required to determine carbon intensity factors used in enterprise-level emission estimation (Habib et al., 2024; Kim et al., 2022; Wood et al.; Roelands et al., 2018; Maynard & Funk; Jain et al., 2024; Balaji et al., 2023a). Current carbon accounting methodologies often depend on costly, resource-intensive ecological and economic daltabases, such as the top-down Environmental Extended Multi-regional Input-output (EE-MRIO) model (Stadler et al., 2018; Guo et al., 2024a) or bottom-up Life Cycle Analysis (LCA) methods (Ayres, 1995). This complexity hinders broader adoption, especially among small and medium enterprises (SMEs), and is exacerbated by a dearth of large-scale benchmark datasets that automate sector classification and carbon factor assignment (Afolabi et al., 2023). The absence of such resources creates a major barrier, limiting GHG reporting largely to organizations with sufficient capital and expertise.

To address these challenges, we propose **ExioNAICS**, the first large-scale benchmark dataset for enterprise-level GHG emission estimation. Our dataset covers over 20,850 enterprises, each mapped to validated North American Industry Classification System (NAICS) codes (Murphy, 1998), and incorporates carbon intensity factors from 166 sectors through the ExioML economic dataset. We additionally formulate the automated emission estimation pipeline as an Information Retrieval (IR) problem and demonstrate the potential of Natural Language Processing (NLP) techniques and Large Language Models (LLMs) in streamlining carbon accounting. With this approach, we achieve 77.93% Top-1 accuracy and 91.33% Top-10 accuracy in industry classification. This study contributes in three key ways: **(1)** it introduces a novel, publicly available benchmark dataset for enterprise-level GHG estimation, **(2)** it applies state-of-the-art NLP models to automate sector classification, and **(3)** it provides a standardized emission estimation pipeline that helps bridge machine learning research with climate science, thereby making carbon accounting more accessible to SMEs.

2 ExioNAICS Emission Estimation Dataset Construction

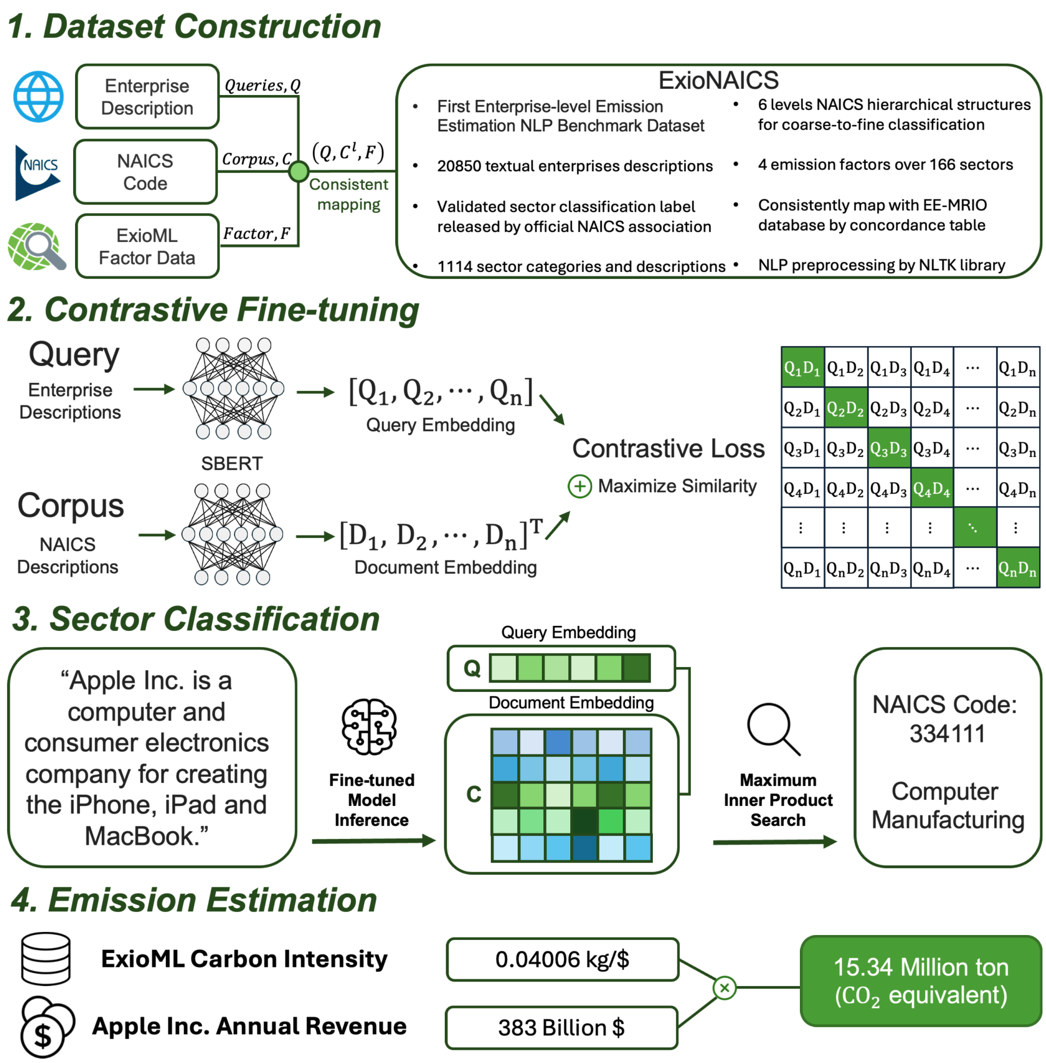

Figure 1: Overview of the ExioNAICS pipeline. (1) *Dataset Construction:* Enterprise descriptions, NAICS codes, and ExioML factor data are consistently mapped into a unified dataset. (2) *Contrastive Fine-Tuning:* We fine-tune SBERT using a contrastive loss to maximize the similarity between matching enterprise–NAICS pairs. (3) *Sector Classification:* After training, queries (enterprise descriptions) are matched with the correct NAICS code via Maximum Inner Product Search in the shared embedding space. (4) *Emission Estimation:* Mapped NAICS codes allow retrieval of sector-specific carbon intensity factors for final GHG emission calculation.

Efforts in expert dataset construction, such as medical image segmentation, often emphasize maintaining inter-annotator consistency (Zhang et al., 2021; Tajbakhsh et al., 2020). Similar challenges appear in NAICS classification: one study found up to 51% label disagreement among annotators in production settings (Balaji et al., 2023b). Recent work has explored using generative AI to create synthetic mappings between NAICS classification and emission factor databases (Dumit et al., 2024).

Building on these insights, **ExioNAICS** integrates enterprise-level and sector-level data using NAICS as a consistent intermediary, as illustrated in Figure 1. We collected more than 20,850 textual descriptions from 13,823 unique representative enterprises across diverse sectors (e.g., Apple, Amazon, Toyota). We then retrieved validated NAICS codes from the official NAICS Association as our ground-truth labels. For emission factors, we use ExioML—an EE-MRIO-based dataset (ExioBase 3.8.2) curated to streamline economic and environmental data for machine learning applications (Guo et al., 2024b; Stadler et al., 2018). This dataset contains carbon intensity factors for 166 sectors, paired with a concordance table to align with NAICS codes.

ExioNAICS preserves the hierarchical NAICS structure, which spans from 2-digit to 6-digit levels. Higher-digit levels correspond to finer-grained sector distinctions, ranging from fewer than 20 classes (2-digit) to over 1,000 classes (6-digit). We define the dataset D as a tuple (Q, C^l, F) for a typical information retrieval task, where Q represents enterprise descriptions, C^l denotes NAICS description corpus at level $l \in \{2, 3, 6\}$, and F contains emission factors from ExioML. Text preprocessing is performed using the NLTK library, including lowercasing, URL removal, and eliminating special characters. Table 1 summarizes the main dataset statistics.

Table 1: Key Statistics of the ExioNAICS Dataset

Metric	Q	C^2	C^3	C^6	F
Unique Class	13,823	20	95	1,114	166
Min Length	6	103	17	10	–
Avg Length	33	515	153	56	–
Max Length	154	846	430	164	–
Data Size $\lvert D \rvert$					20,850

3 Fine-Tuning SBERT via Contrastive Learning

We frame the sector classification problem as an information retrieval task. Let f_θ denote a pretrained Sentence-BERT (SBERT) model parameterized by θ (Reimers, 2019). SBERT follows a dual-tower architecture: one tower encodes the enterprise description (query q), while the other encodes the NAICS description corpus $C = \{d_1, d_2, \ldots, d_n\}$. We identify the most relevant NAICS document d^* for a given enterprise description q via Maximum Inner Product Search (MIPS) based on cosine similarity:

$$d^* = \arg\max_{d \in C} \frac{f_\theta(q) \cdot f_\theta(d)}{\|f_\theta(q)\| \|f_\theta(d)\|}. \tag{1}$$

Rather than using supervised cross-entropy, we fine-tune SBERT in a *contrastive learning* framework (Chen et al.; He et al.; Grill et al., 2020; Radford et al.; Caron et al., 2020), which has been shown to be particularly effective for tasks with sparse or noisy labels (Balaji et al., 2023b). Specifically, we employ the Multiple Negative Ranking (MNR) loss (Henderson et al., 2017):

$$\mathcal{L} = -\frac{1}{n} \sum_{i=1}^{n} \log \left(\frac{\exp(\cos(Q_i, D_i))}{\sum_{j=1}^{n} \exp(\cos(Q_i, D_j))} \right), \tag{2}$$

where n is the batch size, $Q_i = f_\theta(q_i)$ and $D_i = f_\theta(d_i)$ denote embeddings of the i-th query and document, respectively, and $\cos(\cdot, \cdot)$ is the cosine similarity function. Diagonal elements in the similarity matrix $\{S_{ii}\}$ represent positive query–document pairs that we aim to maximize, while off-diagonal elements $\{S_{ij}\}$ for $i \neq j$ are treated as negative pairs.

4 EXPERIMENTAL SETUP

We evaluate various SBERT backbones on classification tasks at NAICS levels 2, 3, and 6, which range from 20 to 1,114 classes. Each subset is split into 80% training, 10% validation, and 10% testing. All models are trained for 100 epochs with the MNR loss, a learning rate of 2×10^{-5}, and a warm-up ratio of 0.1 on an NVIDIA T4 GPU with mixed precision (FP16).

We measure performance using Top-k accuracy (denoted Acc@k):

$$\text{Acc@}k = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}}, \tag{3}$$

where a prediction is considered correct (True Positive, TP) if any ground-truth NAICS code is included among the top-k predictions. False Positives (FP) refer to cases where an incorrect code is predicted in the top-k, and False Negatives (FN) occur when no correct code appears in the top-k.

5 RESULTS

Table 2 reports the Top-1 accuracy of different SBERT backbones at NAICS levels 2, 3, and 6. Among larger models, *all-mpnet-base-v2* achieves the highest accuracy (91.73%, 88.54%, and 82.87%, respectively), but requires longer training (around five hours). MiniLM variants strike a better balance between efficiency and performance; for instance, *all-MiniLM-L12-v2* achieves 91.23%, 87.84%, and 82.67% in about two hours.

Table 2: Top-1 Accuracy (%) of Various SBERT Backbones on NAICS Levels 2, 3, and 6

Backbone	**Pre-Training**	**Size (MB)**	$\mathbf{C^2}$	$\mathbf{C^3}$	$\mathbf{C^6}$
MiniLM$_{L3}$	Paraphrase	**60**	89.73	86.02	77.51
MiniLM$_{L6}$	All	80	91.68	87.93	78.34
MiniLM$_{L6}$	Multi-QA	80	89.14	86.52	79.57
MiniLM$_{L12}$	All	120	91.23	87.84	82.67
Mpnet$_{base}$	All	420	**91.73**	**88.54**	**82.87**

In an ablation study on the most challenging NAICS level 6 task (Table 3), we observe that a zero-shot of *MiniLM-L3-v2* yields only 20.12% Top-1 accuracy, which improves marginally to 21.49% with a five-epoch fine-tuning under cross-entropy loss. By contrast, fine-tuning under a contrastive MNR loss increases Top-1 accuracy dramatically to 76.85%. Further applying NLTK-based text preprocessing yields an additional boost to 77.51%, underscoring the importance of data cleaning.

Table 3: Ablation Study on NAICS Level 6 Classification (MiniLM-L3-v2)

Method	Acc@1	Acc@3	Acc@5	Acc@10
Zero-Shot SBERT	20.12	35.51	43.27	53.52
Cross-Entropy Loss	21.49	36.39	45.58	55.42
Contrastive Loss	76.85	85.42	87.85	90.28
NLTK Preprocessed	**77.51**	**87.37**	**89.22**	**91.33**

6 CONCLUSION

In this work, we introduced **ExioNAICS**, the first large-scale benchmark dataset designed for enterprise-level GHG emission estimation. Our approach integrates over 20,850 enterprise descriptions, validated NAICS codes, and 166 sector-level carbon intensity factors from the ExioML database. By framing the task as an Information Retrieval problem and fine-tuning Sentence-BERT via a contrastive loss, we achieve notable improvements in sector classification, culminating in a Top-1 accuracy of 77.51% and Top-10 accuracy of 91.33% on the most challenging 1,114-class setting. These results underscore the effectiveness of modern NLP techniques in automating carbon accounting pipelines, making them significantly more accessible and cost-effective for enterprises.

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